

Module 6: Multivariable & Constrained Optimization

Ph.D. Math Camp

Partial Differentiation

Partial Differentiation

- ▶ Often, the functions we are interested in are function of more than a single variable. For example, in microeconomics Q , output, is often a function of capital (K) and labor (L).
- ▶ Let's consider a function

$$y = f(x_1, x_2, \dots, x_n)$$

where $x_1, x_2, \dots, x_n$ are independent of each other. If x_1 undergoes a change Δx_1 , while $x_2, \dots, x_n$ remain fixed, there will be a corresponding change in y , Δy .

Partial Differentiation

- ▶ We can express this using our old secant formula

$$\frac{\Delta y}{\Delta x_1} = \frac{f(x_1 + \Delta x_1, x_2, \dots, x_n) - f(x_1, x_2, \dots, x_n)}{\Delta x_1}$$

- ▶ This indicates all other variables will be held constant. We use the following notation $\frac{\partial y}{\partial x_1}$. Read “partial f, partial x one.”
- ▶ Instead of f' we use f_1 or f_x , where the subscript indicates the variable we are taking the partial w.r.t.

Partial Differentiation

For example, the partial derivatives of

$$f(x_1, x_2) = 3x_1^2 + x_1x_2 + 4x_2^2$$

$$f_1 = 6x_1 + x_2$$

$$f_2 = x_1 + 8x_2$$

All normal differentiation rules apply.

Partial Differentiation

Find partials of the following

1. $z = 5x^3 - 3x^2y^2 + 7y^5$

2. $f = xe^{-2xy}$

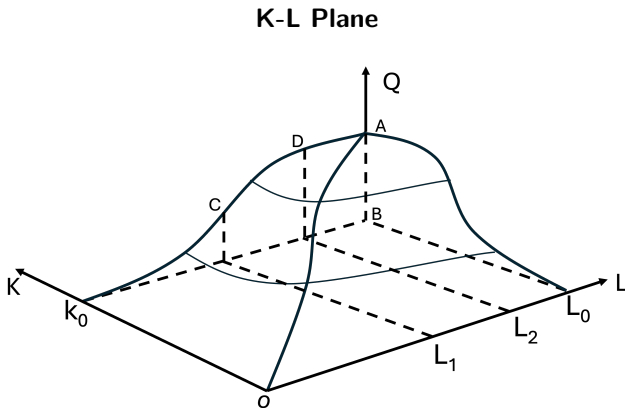
3. $z = 4x^3 + 2x^2y - 7y^5$

Geometric Interpretation of Partial

- ▶ Given that the partial is a type of derivative it has a slope interpretation.
- ▶ Consider $Q = Q(K, L) \implies$ we can define two partials $\frac{\partial Q}{\partial K}$ or Q_K and $\frac{\partial Q}{\partial L}$ or Q_L .
- ▶ Q_K : The rate of change of output with respect to a small change in capital, while labor input is held constant
 $\implies MP_K$ or **marginal product of K**.
- ▶ Similarly, Q_L , MP_L is the **marginal product of labor**.

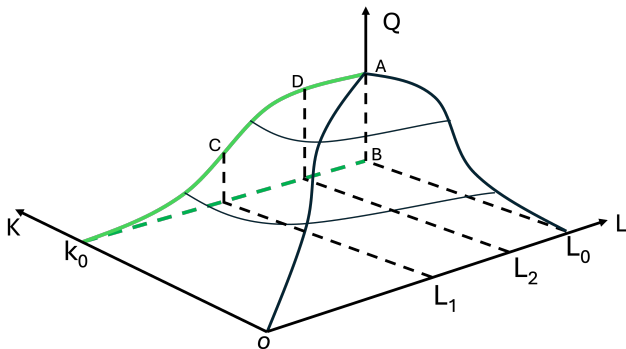
Geometric Interpretation of Partial

- ▶ We can depict $Q = Q(K, L)$ with a production surface.
- ▶ Q is plotted vertically so that for any point (K, L) in the (base) K, L plane the height of the surface will indicate output. The domain should be the non-negative quadrant.



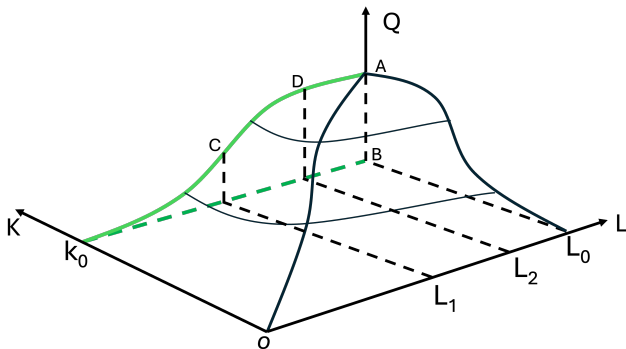
Geometric Interpretation of Partial

- ▶ Suppose we hold K fixed at k_0 and consider variations in the input of L . **By setting $K = k_0$, all points in the domain become irrelevant except those on the k_0B line.**
- ▶ **Only the curve k_0CDA matters.** This curve represents the total product of labor (TP_L) for a fixed amount of $K = k_0$.



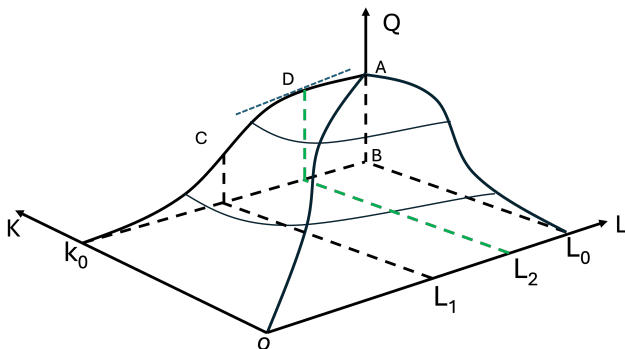
Geometric Interpretation of Partial

- $\implies$ the slope of curve k_0CDA represents the geometric counterpart of Q_L . **k_0CDA is a cross section of the production surface.**
- $\implies$ slope is MP_L . Note that Q_L is a function of L . When $L = L_1$, the value of Q_L is the slope at point c .



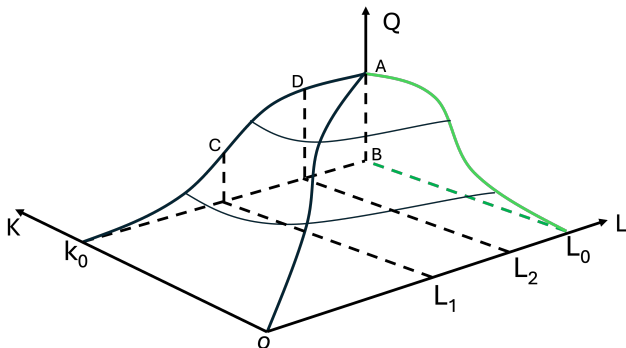
Geometric Interpretation of Partial

- ▶ When $L = L_2$, the relevant point is D.
- ▶ **Why is Q_L also a function of K ?** K can be fixed at various levels. For each value of k , we have a different value of TP_L .



Geometric Interpretation of Partial

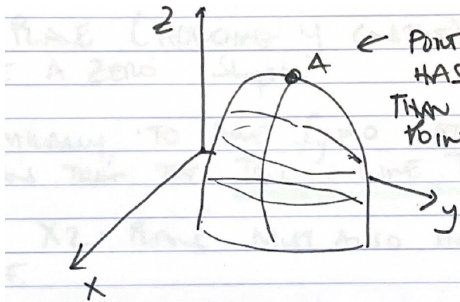
- ▶ We can have a **similar discussion** about Q_K . Suppose we held labor constant at $L_0 \Rightarrow L_0B$ is the relevant line segment.
- ▶ $\Rightarrow L_0A$ is the relevant subset of the production surface and Q_K is the slope of the curve L_0A and is a function of L and K .



Multivariable Optimization

Multivariable Optimization

- ▶ For a function of one choice variable, extreme values are peaks and valleys.
- ▶ When we have two choice variables, the graph $z = f(x, y)$ becomes a surface in 3-D space and our peaks and valleys take on a 3-D form $\Rightarrow$ a dome or a bowl.



- ▶ Point A has the largest z value in its neighborhood.

First-Order Condition

- ▶ A point in the domain of $z = f(x, y)$ are critical points when

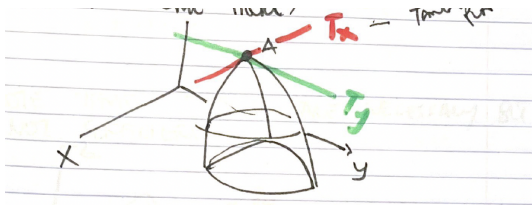
$$f_x = f_y = 0 \quad \text{or} \quad \frac{\partial z}{\partial x} = \frac{\partial z}{\partial y} = 0$$

First-Order Condition

- ▶ A point in the domain of $z = f(x, y)$ are critical points when

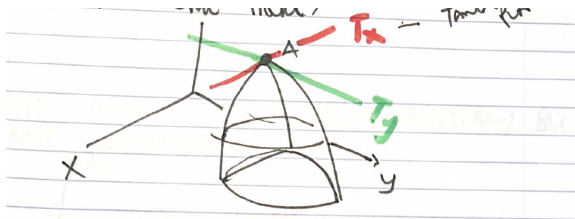
$$f_x = f_y = 0 \quad \text{or} \quad \frac{\partial z}{\partial x} = \frac{\partial z}{\partial y} = 0$$

- ▶ To have $f_x = 0$ at A means tangent line T_x (drawn through A and parallel to xz plane (holding y constant), must have zero slope.



First-Order Condition

- ▶ Similarly, to have $f_y = 0$ at point A means that the tangent line T_y , parallel to yz -plane must also have zero slope.
- ▶ Similar, in case of a minimum.
- ▶ These F.O.C. or necessary conditions: $f_x = 0$, $f_y = 0$



First-Order Condition

Suppose we want to find the points that satisfy F.O.C. for $z = f(x, y) = 3x^2 - xy + 2y^2 - 4x - 7y + 12$

$$f_x = 6x - y - 4 = 0$$

$$f_y = -x + 4y - 7 = 0$$

$$\implies y = 2, x = 1$$

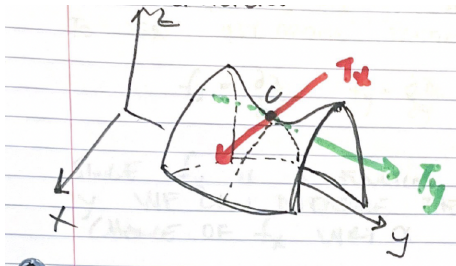
First-Order Condition

- ▶ Note that F.O.C. are necessary but not sufficient.

First-Order Condition

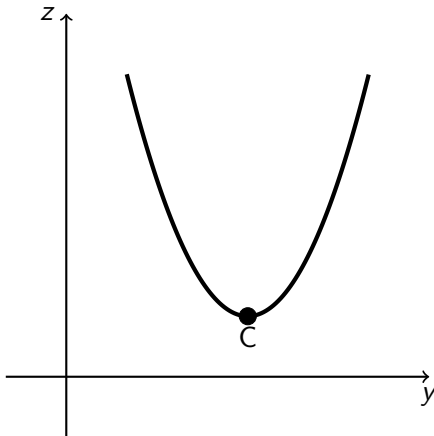
- ▶ Note that F.O.C. are necessary but not sufficient.
- ▶ For instance, at point C, T_x and T_y have zero slopes, but this point does not qualify as an extremum.

Figure: Saddle Point



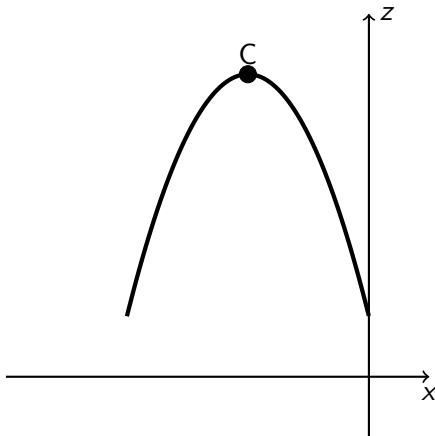
First-Order Condition

- ▶ When viewed against the yz -plane you get a **minimum**.



First-Order Condition

- ▶ When viewed against the xz — plane you get a **maximum**



Second-Order Conditions

- ▶ To develop a sufficient condition we need second-order partial derivatives.
- ▶ Recall that $z = f(x, y)$ can give rise to two first order partials

$$f_x = \frac{\partial z}{\partial x} \quad \text{and} \quad f_y = \frac{\partial z}{\partial y}$$

- ▶ Since f_x is a function of x and y we can measure the rate of change of f_x with respect to x and y .

Second-Order Conditions

- Differentiating with respect to x twice

$$f_{xx} = \frac{\partial}{\partial x} f_x \quad \text{or} \quad \frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x} \right)$$

- Same for y

$$f_{yy} = \frac{\partial}{\partial y} f_y \quad \text{or} \quad \frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial y} \right)$$

Second-Order Conditions

- ▶ We can also obtain cross (or mixed) partial derivatives

$$f_{xy} = \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right)$$

and

$$f_{yx} = \frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x} \right)$$

- ▶ In this case the partial measures the rate of change of one first order partial with respect to the “other” variable.
- ▶ Even though f_{xy} and f_{yx} have been separately defined, they **will have identical values** (as long as the cross partials are continuous)– known as *Young’s theorem*.
- ▶ Also works for functions with 3 or more variables. If $z = g(u, v, w) \implies g_{uv} = g_{vu}; g_{vw} = g_{wv}$.

Second-Order Conditions

Find $f_{xx}, f_{xy}, f_{yx}, f_{yy}$

1. $z = f(x, y) = x^3 + 5xy - y^2$

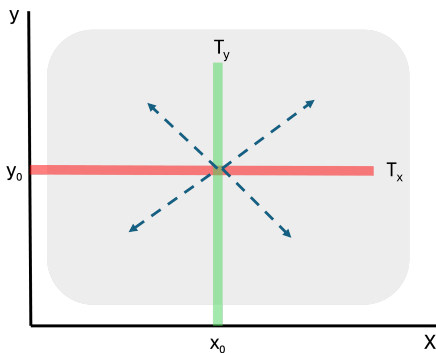
2. $z = f(x, y) = x^2 e^{-y}$

Interpreting S.O.C.

- ▶ Let $D = f_{xx}f_{yy} - (f_{xy})^2$.
 - ▶ We will see a max if $f_{xx} < 0, f_{yy} < 0$ and $D > 0 \implies f_{xx}f_{yy} > (f_{xy})^2$
 - ▶ We get a minimum if $f_{xx} > 0, f_{yy} > 0$ and $D > 0 \implies f_{xx}f_{yy} > (f_{xy})^2$
- ▶ The S.O.C. depends on f_{xx} and f_{yy} which have to do with the surface configuration around point A.

Interpreting S.O.C.

- We consider the cross partials as well. Let's take a bird's eye view.



- The cross partial ensures that the surface that is tangent will yield the same type of configuration (i.e. hill or valley) in any direction (NE, SW, etc.)

Interpreting S.O.C.

Let's summarize with a table

Condition	Max	Min
F.O.C.	$f_x = f_y = 0$	$f_x = f_y = 0$
S.O.C.	$f_{xx}, f_{yy} < 0$ and $f_{xx}f_{yy} > f_{xy}^2$	$f_{xx}, f_{yy} > 0$ and $f_{xx}f_{yy} > f_{xy}^2$

- Note that S.O.C are sufficient but not necessary (e.g., $D = 0$).

First and S.O.C.

Find extreme values and identify as min or max for the following

1. $z = 8x^3 + 2xy - 3x^2 + y^2 + 1$

2. $z = x + 2e^y - e^x - e^{2y}$

First and S.O.C.

1. $z = 8x^3 + 2xy - 3x^2 + y^2 + 1$

$$f_x : 24x^2 + 2y - 6x = 0$$

$$f_y : 2x + 2y = 0$$

$$\implies x = -y \implies 24x^2 + 2(-x) - 6x = 0$$

$$\implies 24x^2 - 2x - 6x = 0 \implies 24x^2 - 8x = 0$$

$$\implies x(24x - 8) = 0$$

$$\implies x_1^* = 0, y_1^* = 0, \quad x_2^* = \frac{1}{3}, y_2^* = -\frac{1}{3}$$

First and S.O.C.

SOC at $x = 0, y = 0$

$$f_{xx} = 48x - 6 \implies f_{xx} \Big|_{x=0} = -6$$

$$f_{yy} = 2$$

$$f_{xy} = f_{yx} = 2$$

Since $f_{xx}f_{yy} < f_{xy}^2 \implies$ a saddle point $\implies$ surface curls up in one dimension but down in another.

First and S.O.C.

$$\text{SOC at } x = \frac{1}{3}y = \frac{-1}{3}$$

$$f_{xx} = 48x - 6 \implies \left|_{x=\frac{1}{3}} = 16 - 6 = 10\right.$$

$$f_{yy} = 2$$

$$f_{xy} = f_{yx} = 2$$

Since $f_{xx}f_{yy} > f_{xy}^2$ or $10 \times 2 > (2)^2 \implies$ a **minimum**.

Determinant Test for Sign Definiteness

Sign Definiteness

- ▶ The Second order sufficient condition for extremum stipulates it be **definitely positive** (for a minimum) and **definitely negative** (for a maximum).

Sign Definiteness

- ▶ We can rewrite the second order conditions as follows
 - ▶ Positive definite if $f_{xx} > 0$ and $\begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix}$ is greater than zero.
 - ▶ Negative definite if $f_{xx} < 0$ and $\begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix}$ is greater than zero.
 - ▶ Since f_{xx} and f_{yy} need to have the same sign we get the condition from before.
- ▶ $\begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix}$ is known as the discriminant. This is a determinant with second order partial derivatives as its elements.
- ▶ We call this the **Hessian Determinant** (or Hessian).

Sign Definiteness

- ▶ The **Hessian Determinant** (or Hessian).

$$|H| = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} = f_{xx}f_{yy} - f_{xy}^2$$

Objective Functions with 2+ Variables

- ▶ We can no longer graph a function when there are more than two choice variables.
- ▶ However, we can still speak of hyper-surfaces that may be tangent in a way to peaks and domes in this hyperdimensional space.
- ▶ Consider $f(x_1, x_2, x_3) \implies$ 1st partials f_1, f_2, f_3 and 2nd partials $f_{ij} = \frac{\partial^2 z}{\partial x_i \partial x_j}$ for $i, j = 1, 2, 3$.
- ▶ By virtue of Young's theorem $f_{ij} = f_{ji}$.
- ▶ FOC $\implies f_1 = f_2 = f_3 = 0$. Just as in the two-variable case we need that 1st order partial derivative to be zero.

Objective Functions with 2+ Variables

- ▶ For second order condition, if at a stationary value of z we find that the function (there) is positive definite $\implies$ that value of z is a minimum.
- ▶ Analogously, the negative definiteness is a sufficient condition for a value to be a max.
- ▶ To determine this, we set up our Hessian of second partials

$$|H| = \begin{vmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{vmatrix}$$

- ▶ We can then find the **principal minors**:

$$|H_1| = f_{11}, \quad |H_2| = \begin{vmatrix} f_{11} & f_{12} \\ f_{21} & f_{22} \end{vmatrix}, \quad |H_3| = |H| \quad (1)$$

Second Order Conditions

- ▶ Note that all leading principal minors must be evaluated at the stationary points where $f_1 = f_2 = f_3 = 0$.
 - ▶ Z^* is a maximum if $|H_1| < 0; |H_2| > 0; |H_3| < 0 \implies$ negative definiteness
 - ▶ Z^* is a minimum if $|H_1| > 0; |H_2| > 0; |H_3| > 0 \implies$ positive definiteness

Find extreme values of following and determine if they are min or max.

1. $z = 2x_1^2 + x_1x_2 + 4x_2^2 + x_1x_3 + x_3^2 + 2$

2. $z = -x_1^3 + 3x_1x_3 + 2x_2 - x_2^2 - 3x_3^2$

Hint: this one has two stationary points. At $(0, 1, 0)$ the leading-principal-minor test is inconclusive ($|H_1| = 0$) — focus on classifying the other stationary point.

First and Second Order Conditions

Summary

Condition	Max	Min
F.O.C.	$f_1 = \dots = f_n = 0$	$f_1 = \dots = f_n = 0$
S.O.C.	$ H_n > 0$ (< 0) for n even (odd)	$ H_n > 0$ for all n

Constrained Optimization

Effects of a Constraint

- ▶ We just discussed finding relative extrema of an objective function with 2 or more variables.
- ▶ However, in that discussion the decision made regarding one variable does not impinge upon the choices of the remaining variables $\rightarrow$ *constrained optimization*.

Effects of a Constraint

- ▶ Let's consider a utility function $U = x_1x_2 + 2x_1$. Where consumption (choice) of the number of x_1 and x_2 increase wellbeing, U .
- ▶ U_1 and U_2 are the partial derivatives and known as marginal utilities.

$$U_1 = \frac{\partial U}{\partial x_1} \quad U_2 = \frac{\partial U}{\partial x_2}$$

$$\implies U_1 = x_2 + 2, \quad U_2 = x_1$$

- ▶ Since they are both positive (e.g., as x_1 and x_2 increase $\implies U$ increases). This would imply purchasing an infinite number of both goods.
- ▶ Realistically, a consumer would face a **budget constraint**.

Effects of a Constraint

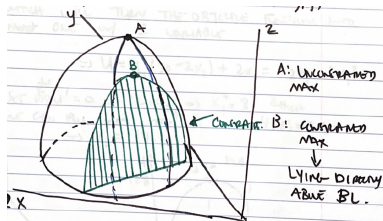
- ▶ Suppose the consumer allocates \$60 on two goods with prices $p_1 = 4$ and $p_2 = 2$, then their budget line can be expressed as

$$4x_1 + 2x_2 = 60$$

- ▶ Once a BL is introduced, the domain is immediately reduced to the set of points lying on (or below) the budget line.
- ▶ Only the subset of the utility surface lying directly above the budget constraint line will be relevant.

Effects of a Constraint

- Generally, for a function $z = f(x, y)$



- Point A would be the unconstrained max
- Point B would be the constrained max, which lies above the budget line.
- Constrained max is usually $\leq$ unconstrained max (although usually less).
- Note that if we added another intersecting line $\implies$ solution is a single point $\implies$ usually there would be fewer constraints than choice variables.

Finding Stationary Values

- ▶ Consider $\text{Max } U = x_1x_2 + 2x_1 \quad \text{s.t.} \quad 4x_1 + 2x_2 = 60.$

Finding Stationary Values

- ▶ Consider $\text{Max } U = x_1x_2 + 2x_1 \quad \text{s.t.} \quad 4x_1 + 2x_2 = 60$.
- ▶ We can substitute $x_2 = 30 - 2x_1$ into the $U(x_1, x_2)$. This turns the objective function into one with 1 variable.

$$\implies U = x_1(30 - 2x_1) + 2x_1 = 32x_1 - 2x_1^2 \quad (2)$$

Finding Stationary Values

- ▶ Consider Max $U = x_1x_2 + 2x_1$ s.t. $4x_1 + 2x_2 = 60$.
- ▶ We can substitute $x_2 = 30 - 2x_1$ into the $U(x_1, x_2)$. This turns the objective function into one with 1 variable.

$$\implies U = x_1(30 - 2x_1) + 2x_1 = 32x_1 - 2x_1^2 \quad (2)$$

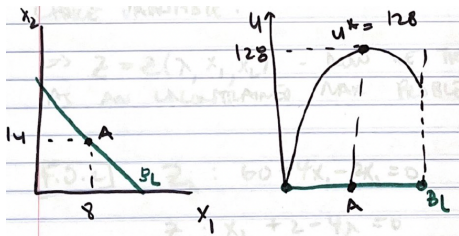
- ▶ Set $\frac{dU}{dx} = U' = 0 \implies 32 - 4x_1 = 0 \implies x_1^* = 8$.

Finding Stationary Values

- ▶ Consider $\text{Max } U = x_1x_2 + 2x_1 \quad \text{s.t. } 4x_1 + 2x_2 = 60$.
- ▶ We can substitute $x_2 = 30 - 2x_1$ into the $U(x_1, x_2)$. This turns the objective function into one with 1 variable.

$$\Rightarrow U = x_1(30 - 2x_1) + 2x_1 = 32x_1 - 2x_1^2 \quad (2)$$

- ▶ Set $\frac{dU}{dx} = U' = 0 \Rightarrow 32 - 4x_1 = 0 \Rightarrow x_1^* = 8$.
- ▶ We can then plug this into the BL
 $\Rightarrow 4(8) + 2x_2 = 60 \Rightarrow x_2 = 14$.



Finding Stationary Values

- ▶ Since $\frac{d^2U}{dx_1^2} = -4 < 0 \implies$ a max.
- ▶ If the constraint is more complicated or there are several, this method can become difficult.

Lagrange-Multiplier Method

Lagrangian

- ▶ Using

$$\text{Max } U = x_1x_2 + 2x_1 \quad \text{s.t. } 4x_1 + 2x_2 = 60$$

- ▶ We can write down the lagrangian function (note how we are adding zero)

$$Z = x_1x_2 + 2x_1 + \lambda(60 - 4x_1 - 2x_2)$$

- ▶ We can treat λ as an additional choice variable.
- ▶ We can now treat $Z = Z(\lambda, x_1, x_2)$ as an unconstrained maximization problem.

Lagrangian

$$Z = x_1x_2 + 2x_1 + \lambda(60 - 4x_1 - 2x_2)$$

► **FOC:**

$$Z_\lambda : 60 - 4x_1 - 2x_2 = 0 \quad (1)$$

$$Z_1 : x_2 + 2 - 4\lambda = 0 \quad (2)$$

$$Z_2 : x_1 - 2\lambda = 0 \quad (3)$$

- Z_λ guarantees the satisfaction of the constraint even though we treated this as an unconstrained max problem.

Lagrangian

FOC:

$$Z_{\lambda} : 60 - 4x_1 - 2x_2 = 0 \quad (1)$$

$$Z_1 : x_2 + 2 - 4\lambda = 0 \quad (2)$$

$$Z_2 : x_1 - 2\lambda = 0 \quad (3)$$

- ▶ The Lagrange method allows us to obtain an unconstrained extremum U^* . We can solve for the critical values:

$$\text{From (3)} \implies x_1 = 2\lambda$$

$$\text{From (2)} \implies x_2 = 4\lambda - 2$$

$$\text{From (1)} \implies 60 - 4(2\lambda) - 2(4\lambda - 2) = 0$$

$$\implies 60 - 8\lambda - 8\lambda + 4 = 0 \implies 64 = 16\lambda \implies \lambda = 4$$

$$\implies x_1^* = 8, x_2^* = 14.$$

Lagrangian

- Generally, for a function $z = f(x, y)$ s.t. $g(x, y) = c$, c is a constant

$$\mathcal{L} = f(x, y) + \lambda[c - g(x, y)]$$

$\Rightarrow$

$$\mathcal{L}_\lambda : c - g(x, y) = 0 \quad (1)$$

$$\mathcal{L}_x : f_x - \lambda g_x = 0 \quad (2)$$

$$\mathcal{L}_y : f_y - \lambda g_y = 0 \quad (3)$$

- The ingenuity here is that $c - g(x, y)$ is zero so max at $\mathcal{L}$ will be the same as that at Z s.t. $g(x, y) = c$. Since BL is considered in FOC.

Lagrangian

- ▶ Use Lagrangian to find critical points for $z = xy$ s.t. $x + y = 6$

Interpretation of Lagrange Multiplier

- ▶ It turns out that λ^* is a measure of the sensitivity of Z^* to changes in the constraint.
- ▶ Given the constraint in Z_λ (FOC) we can think of all the solutions as a function of c .

Interpretation of Lagrange Multiplier

- From $Z_\lambda : c - g(x, y) = 0$

$\implies$ the optimal solutions are $\lambda^* = \lambda^*(c), x^* = x^*(c), y^* = y^*(c)$

$$\implies Z^* = f(x^*, y^*) + \lambda^*[c - g(x^*, y^*)]$$

If we differentiate Z^* w.r.t. $c \implies$

$$\frac{dZ^*}{dc} = f_x \frac{dx^*}{dc} + f_y \frac{dy^*}{dc} + \frac{d\lambda^*}{dc} [c - g(x^*, y^*)] + \lambda^* (1 - g_x \frac{dx^*}{dc} - g_y \frac{dy^*}{dc})$$

- Now, combine like terms

$$f_x \frac{dx^*}{dc} - \lambda^* g_x \frac{dx^*}{dc} + f_y \frac{dy^*}{dc} - \lambda^* g_y \frac{dy^*}{dc} + \frac{d\lambda^*}{dc} [c - g(x^*, y^*)] + \lambda^*$$

- We get

$$(f_x - \lambda^* g_x) \frac{dx^*}{dc} + (f_y - \lambda^* g_y) \frac{dy^*}{dc} + \frac{d\lambda^*}{dc} [c - g(x^*, y^*)] + \lambda^*$$

Interpretation of Lagrange Multiplier

- We get

$$(f_x - \lambda^* g_x) \frac{dx^*}{dc} + (f_y - \lambda^* g_y) \frac{dy^*}{dc} + \frac{d\lambda^*}{dc} [c - g(x^*, y^*)] + \lambda^*$$

- The first three terms are equal to zero from FOC.
 $\implies \frac{dZ^*}{dc} = \lambda^*.$
- This means that the effect of a change in the constraint via c on optimal value z^* is λ^* .

n-Variable and Multiconstraint Cases

- Now suppose we have

$$Z = f(x_1, x_2, \dots, x_n) \quad \text{s.t.} \quad g(x_1, \dots, x_n) = c$$

$$\mathcal{L} = f(x_1, x_2, \dots, x_n) + \lambda[c - g(x_1, \dots, x_n)]$$

$$\mathcal{L}_\lambda : c - g(x_1, \dots, x_n) = 0$$

$$\mathcal{L}_1 : f_1 - \lambda g_1 = 0$$

$$\mathcal{L}_2 : f_2 - \lambda g_2 = 0$$

.....

$$\mathcal{L}_n : f_n - \lambda g_n = 0$$

n-Variable and Multiconstraint Cases

- If you have two constraints

$\mathcal{L} = f(x_1, x_2, \dots, x_n) + \lambda[c - g(x_1, \dots, x_n)] + \mu[d - h(x_1, \dots, x_n)]$
you add $\mathcal{L}_\mu$ to first order conditions. Solve for all x_i^* and λ^*, μ^* .

Second Order Conditions

- ▶ Now we must adjust what we did for the unconstrained case. Before we used the Hessian now we need a *bordered Hessian* which we write as follows:

$$|\bar{H}| = \begin{vmatrix} 0 & g_x & g_y \\ g_x & Z_{xx} & Z_{xy} \\ g_y & Z_{yx} & Z_{yy} \end{vmatrix}$$

- ▶ If $|\bar{H}| < 0 \implies d^2z$ **positive definite subject to the constraint** $\implies$ **a minimum.**
- ▶ If $|\bar{H}| > 0 \implies d^2z$ **negative definite subject to the constraint** $\implies$ **a maximum.**
- ▶ Note that all derivatives in $|\bar{H}|$ have to be evaluated at the critical values.

Second Order Conditions

Find critical points and whether they are min or max of
 $Z = xy$ s.t. $x + 2y = 2$.

n-Variable Case

- ▶ When the objective function takes the form

$$Z = f(x_1, x_2, \dots, x_n) \quad \text{s.t.} \quad g(x_1, \dots, x_n) = c$$

- ▶ The condition for positive or negative definiteness involves a bordered hessian.
- ▶ We can express this in terms of **bordered leading principal minors**.

$$|\bar{H}| = \begin{vmatrix} 0 & g_1 & g_2 & \cdots & g_n \\ g_1 & Z_{11} & Z_{12} & \cdots & Z_{1n} \\ g_2 & Z_{21} & Z_{22} & \cdots & Z_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ g_n & Z_{n1} & Z_{n2} & \cdots & Z_{nn} \end{vmatrix}$$

- Its bordered leading principal minors can be defined as

$$|\bar{H}_2| = \begin{vmatrix} 0 & g_1 & g_2 \\ g_1 & Z_{11} & Z_{12} \\ g_2 & Z_{21} & Z_{22} \end{vmatrix}, \quad |\bar{H}_3| = \begin{vmatrix} 0 & g_1 & g_2 & g_3 \\ g_1 & Z_{11} & Z_{12} & Z_{13} \\ g_2 & Z_{21} & Z_{22} & Z_{23} \\ g_3 & Z_{31} & Z_{32} & Z_{33} \end{vmatrix}$$

- The last being $|\bar{H}_n| = |\bar{H}|$.
- We have **positive definiteness (a minimum)** if $|\bar{H}_2|, |\bar{H}_3|, \dots, |\bar{H}_n| < 0$
- We have **negative definiteness (a maximum)** if $|\bar{H}_2| > 0, |\bar{H}_3| < 0, |\bar{H}_4| > 0, \dots$

First and Second Order Conditions

Summary

Condition	Max	Min
F.O.C.	$Z_1 = \dots = Z_n = 0$	$Z_1 = \dots = Z_n = 0$
S.O.C.	$ H_2 > 0, H_3 < 0, H_4 > 0, \dots, (-1)^n H_n > 0$	$ H_n < 0$ for all n

Utility Max and Consumer Demand

Utility Max

- ▶ A classic problem in microeconomics supposes you have a utility function that depends on two goods x, y whose prices p_x and p_y are exogenously determined.
- ▶ Suppose a consumer has budget income I . The problem can be stated as follows

$$\max_{x,y} U = u(x, y) \quad \text{s.t.} \quad p_x x + p_y y = I$$

Utility Max

Set up first order condition using Lagrangian

$$L = u(x, y) + \lambda[I - p_x x - p_y y]$$

$$\mathcal{L}_\lambda : I - p_x x - p_y y = 0 \quad (1)$$

$$\mathcal{L}_x : U_x - \lambda p_x = 0 \quad (2)$$

$$\mathcal{L}_y : U_y - \lambda p_y = 0 \quad (3)$$

- ▶ Using (2) and (3) we get $\frac{U_x}{p_x} = \frac{U_y}{p_y} = \lambda$
- ▶ Classic result from consumer theory: In order for a consumer to max utility, consumers must equalize the ratio of marginal utility to price
- ▶ The additional benefit per dollar for one good equals the additional benefit per dollar of the other good.

Utility Max

- ▶ As before, $\lambda^* = \frac{\partial U^*}{\partial I} \Rightarrow$ the marginal utility of an additional dollar (i.e., relaxing the constraint).

Aside: Total Derivatives

Let's take an aside and consider total derivatives

- ▶ Consider a function $y = f(x, w)$ where $x = g(w)$. Thus, $y = f(g(w), w)$
- ▶ The three variables y , x , and w are all related.
- ▶ w affects y through two separate channels
 - ▶ *Indirectly* via the function g and then f , $f(\mathbf{g}(\mathbf{w}), w)$
 - ▶ *Directly*, via the function f .
- ▶ The direct effect can be represented with f_w or $\frac{\partial y}{\partial w}$
- ▶ The indirect effect can be expressed by the product $f_x \frac{dx}{dw}$ (using chain rule) since it is a composite function.

Aside: Total Derivatives

- ▶ If we add up the two effects we get the **total derivative**

$$\frac{dy}{dw} = f_x \frac{dx}{dw} + f_w$$

- ▶ We can also multiply both sides by dw

$$dy = f_x dx + f_w dw \Rightarrow \text{total differential}$$

Aside: Total Derivatives

Let's find the total derivative of $y = f(x, w) = 3x - w^2$

Aside: Total Derivatives

Let's find the total derivative of $y = f(x, w) = 3x - w^2$ where $x = g(w) = 2w^2 + w$

- ▶ $\frac{dy}{dw} = f_x \frac{dx}{dw} + f_w$
- ▶ $f_x = 3, \frac{dx}{dw} = (4w + 1), f_w = -2w$
- ▶ $\implies \frac{dy}{dw} = 3(4w + 1) + (-2w)$

Utility Max

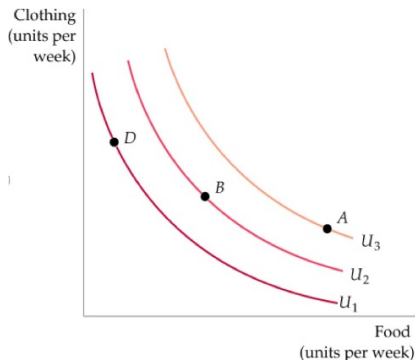
- ▶ Given that we can express the utility function mathematically we can plot a picture— an *indifference curve* (IC)
- ▶ ICs are defined as the locus of the combinations of x and y that will yield a constant level of U .
- ▶ On an IC, $\rightarrow du = U_x dx + U_y dy = 0$

$$\Rightarrow \frac{dy}{dx} = \frac{-U_x}{U_y}$$

- ▶ $\frac{dy}{dx} = \frac{-U_x}{U_y}$ is the slope of an indifference curve

Utility Max

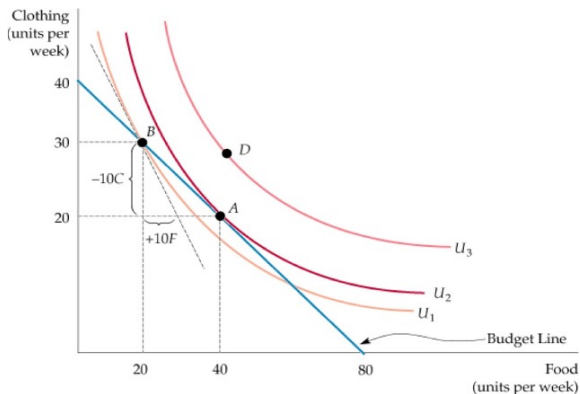
- ▶ We call the slope of an IC, the marginal rate of substitution (MRS).
- ▶ The MRS measures the value that an individual places on an extra unit of a good in terms of another.
- ▶ Which curve is preferred by a consumer?



Utility Max

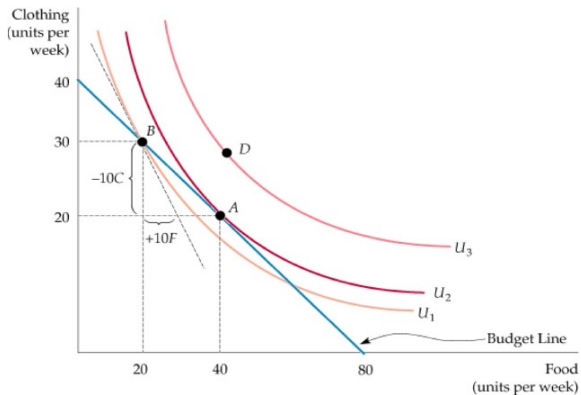
- We can also plot the budget line $p_x x + p_y y = I$ which can be rewritten as

$$y = \frac{I}{p_y} - \frac{p_x}{p_y}x$$



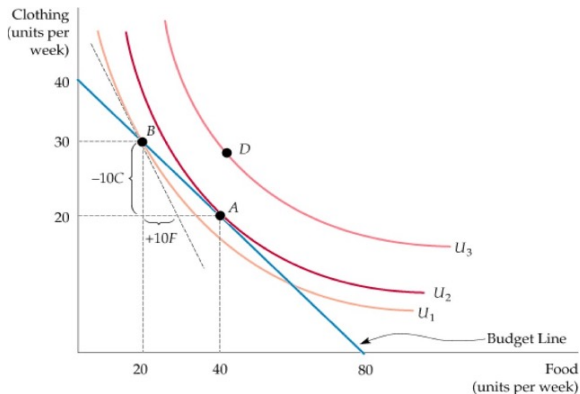
Utility Max

- Which is the most preferred point? A , B , or D ?



Utility Max

- ▶ Which is the most preferred point? A , B , or D ?
- ▶ What is the relationship between the slope of the budget line and the marginal rate of substitution at point B ?

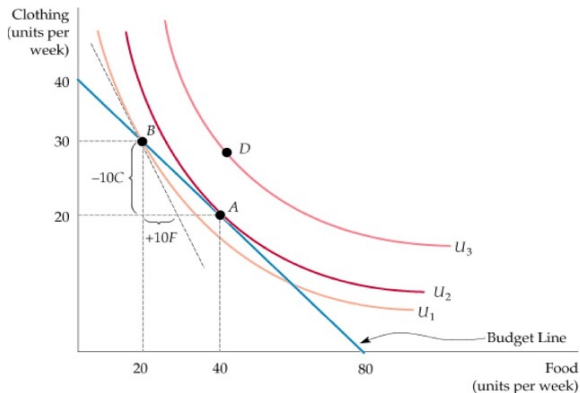


Utility Max

- To maximize utility, a consumer allocates their budget such that the slope of the budget line equals the slope of the ICs.

$$\frac{U_x}{U_y} = \frac{p_x}{p_y}$$

- Point A in this case.



Second Order condition

- If the bordered hessian in this instance is positive $\implies$ a max

$$|\bar{H}| = \begin{vmatrix} 0 & p_x & p_y \\ p_x & U_{xx} & U_{xy} \\ p_y & U_{yx} & U_{yy} \end{vmatrix}$$

$$= 2p_x p_y U_{xy} - p_y^2 U_{xx} - p_x^2 U_{yy} > 0 \implies \text{negative definite}$$